

Tran Thanh Tuan

Full-stack Developer

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PROFILE

Full-stack developer with 4 years on production SaaS, mostly CRM, solar energy and accounting platforms, for US and Vietnamese teams. My work is mainly backend, NestJS and Node.js with PostgreSQL and MongoDB, and I usually own the calculation-heavy part of a product, from the data model through the API to the UI on top of it.

TECHNICAL SKILLS

Languages	TypeScript, JavaScript
Backend	NestJS, Express, Node.js, BullMQ, REST, Zod
Frontend	React, TanStack Query, Zustand, Redux, MUI, Tailwind, Vite
Databases	PostgreSQL (Prisma, Drizzle), MongoDB (Mongoose), Redis
Cloud & CI	AWS, Docker, GitHub Actions
Testing	Jest, Mocha

WORK EXPERIENCE

Circuit Technologies · Full-stack Developer (Freelance)

Jan 2025 – Apr 2026

Solar quoting, financing and commercial bidding platforms, US market

- **Main author of the lease/TPO pricing engine**, backend through to the quoting UI. Rate matrix in MongoDB with **banded lookup**, pricing logic as pure functions unit tested in isolation, and a **pricing snapshot per quote** so quotes stay reproducible after a rate change.
- Built the **commercial bidding workflow** shared by internal staff and external partner companies: **row-level scoping** per partner account, a **cron-driven state machine** on submission deadlines, and email fan-out on award or decline.
- Created a **NestJS caching proxy** over a metered utility-rate API: **two-tier cache** (in-process LRU over MongoDB with a TTL index), cache level switchable at runtime, **fail-open** on cache errors.
- Built the **multi-tenant configuration layer** of a greenfield quoting platform: per-organisation **feature flags** across production and sandbox environments, and compound indexes behind the zip-to-utility and tariff lookups.

NestJS, MongoDB, PostgreSQL/Drizzle · React, MUI, Zustand, Redux · Meteor

Rental Property Management Platform · Full-stack Developer (Freelance)

2024 – Present

Production system managing ~400 rental units

- **Built the accounting reporting layer** over a double-entry ledger. Monthly figures come from MongoDB **aggregation pipelines** while a period is open, then **freeze into an immutable snapshot on close**, with each month opening from the previous closed one, so editing historical data cannot silently change a report that was already signed off.
- Bank-statement ingestion for several Vietnamese banks, **one parser each behind a shared normaliser**, so a new bank format lands in one file. Statements carry no timestamps, so **intra-day order is rebuilt by chaining each row against its running balance**, and a file that will not reconcile is rejected naming the offending rows.
- Cut reconciliation query cost with a **compound index** on the transaction ledger and **partial indexes across ten receipt collections**, scoped to records carrying a bank-transaction link.

Express, MongoDB/Mongoose · Vue 2 · GitHub Actions

Designveloper · Full-stack Developer

Aug 2022 – Oct 2024

Primary client: CRM and quoting platform for a US residential solar provider, from sales quoting through to enrolment in utility grid-services programs, including an 80 MW virtual power plant

- Built the **battery charge/discharge scheduling algorithm**, a time-of-use optimiser deciding hour by hour whether to store or sell energy. Rewrote it for California's NEM 3.0 rule change; still in production three years later.
- Built the energy-profile engine on top of it: **8,760 hourly values per year** into consumption charts, pre- and post-install bill estimates and tariff comparisons.
- Rebuilt the **credit-qualification flow**, decomposing one service around a third-party credit gateway into four, with structured error handling and a persisted event history.
- **Introduced unit testing to a codebase that had none**: 27 test files, including retrofitting coverage onto an untested utility-enrollment pipeline.
- On a second client product, a licensing tool: built OAuth 2.0 and OTP email authentication and set up its GitHub Actions pipeline.

NestJS, MongoDB/Mongoose · React, Redux · Meteor

PROJECTS

HomeHub, personal finance platform, built and operated solo

- **Transactional outbox** on bank webhooks: transaction, balance update and outbox row commit together, a relay drains the outbox into **BullMQ**, and LLM classification runs off the request path.
- **Idempotent consumers** keyed on the outbox event id, so redelivery is a no-op, with a **dead-letter queue** retained for replay.
- Built an **AI agent** for personal finance, using **tool calling** to read accounts and spending and to record transactions. Model output is validated before anything is saved, and writes can be undone.

NestJS, PostgreSQL/Prisma, BullMQ, Redis · React, Vite, TanStack Query, Tailwind · Docker

EDUCATION

Ho Chi Minh University of Technology · Bachelor of Engineering, Computer Science · GPA 8.78/10

Aug 2020 – Dec 2023

CERTIFICATIONS

- AWS Certified AI Practitioner (AIF-C01)